

Project 5: Shiny Dashboard

STAT 133, Spring 2026

About

The purpose of this assignment is to create a **Quarto Dashboard with Shiny** to visualize 2019 crash data in California. In particular, this project will give you a chance to put in practice some of the geospatial concepts covered in weeks 12-13. This assignment is also an excellent opportunity to use a bit of regular expressions, as well as functions for date-time data.

All files related to this assignment are available in bCourses (see *Files/projects/project5*).

<https://bcourses.berkeley.edu/courses/1551809/files/folder/projects/project5>

- `project5.pdf`: instructions
- `crashes_geospatial_2019.csv`: data CSV file
- `Codebook-Crash-Geospatial.pdf`: data documentation (i.e. data dictionary)
- `dashboard-template-local-sidebars.qmd`: template shiny dashboard with local sidebars (i.e. a sidebar for each tab)
- `dashboard-template-global-sidebar.qmd`: template shiny dashboard with a global sidebar (i.e. one sidebar for all tabs)
- `storm-tracks-demo.qmd`: storm tracks demo dashboard (for inspiration/guidance purposes)

General Instructions

- Use a Quarto Document (`.qmd` file) for your Project 5. Please choose one of the template files provided in bCourses, folder `project5/`.
- Save this file as `project5.qmd`.
- Reminder: the YAML header of your `qmd` file must contain the following fields:
 - `format: dashboard`
 - `server: shiny`
- Publish your shiny dashboard to **shinyapps.io**.
- As usual, assume that your published dashboard will be seen by a potential employer, boss, client/customer, admissions committee, or something like that. In other words, make sure that the layout of your dashboard is user friendly, graphical displays are presentation-quality, etc.
- To receive full credit, please submit the following to bCourses (proj5 assignment):
 - `qmd` file
 - (URL) link to your **shinyapps.io** published dashboard.

1 Data

The data for this project is similar to the ones used in Projects 2 and 4. Specifically, the data involves crashes in California during 2019. This time, however, the data includes geospatial information such as longitude, latitude, city, and county.

<https://bcourses.berkeley.edu/courses/1551809/files/folder/projects/project5>

1.1 CSV File

You can find the data, in **CSV** format, in the file `crashes_geospatial_2019.csv` located in bCourses (see `Files/projects/project5`):

We recommend importing the CSV file in R using the `read_csv()` function from the *tidyverse* package "`readr`".

1.2 Data Dictionary

Documentation about the data columns is in the file `Codebook-Crash-Geospatial.pdf` located in bCourses (see `Files/projects/project5`)

2 Shiny Dashboard

Dashboard

The goal is to create a shiny dashboard containing 2 tabs.

Tab 1: to display a web interactive map—via leaflet—that allows the user to visualize and explore the location of certain types of crashes in a given California county.

Tab 2: to display at least one interactive graphic to visualize some sort of time trend in the data.

2.1 Template Files

To help you get started with this project we are providing two template dashboard files (see bCourses `Files/projects/project5`). You only have to choose one of these templates.

- `dashboard-template-global-sidebar.qmd`: this is a template shiny dashboard with a global sidebar (i.e. one sidebar for all tabs).
- `dashboard-template-local-sidebars.qmd`: this is a template shiny dashboard with local sidebars (i.e. a sidebar for each tab).

2.2 Demo File (for inspiration or guidance)

In addition to the template files we are also providing a sample demo `storm-tracks.qmd` available in bCourses (see `Files/projects/project5`). This dashboard allows the user to visualize **storm tracks** of tropical cyclones in the North Atlantic. This demo is provided as an additional resource in case you want to get some inspiration or guidance on how to code some components for your own dashboard.

<https://gastonsanchez.shinyapps.io/storm-tracks/>

2.3 Dashboard Layout

The figure below shows the general layout structure of your shiny dashboard. BTW: This is the layout provided in the template files.

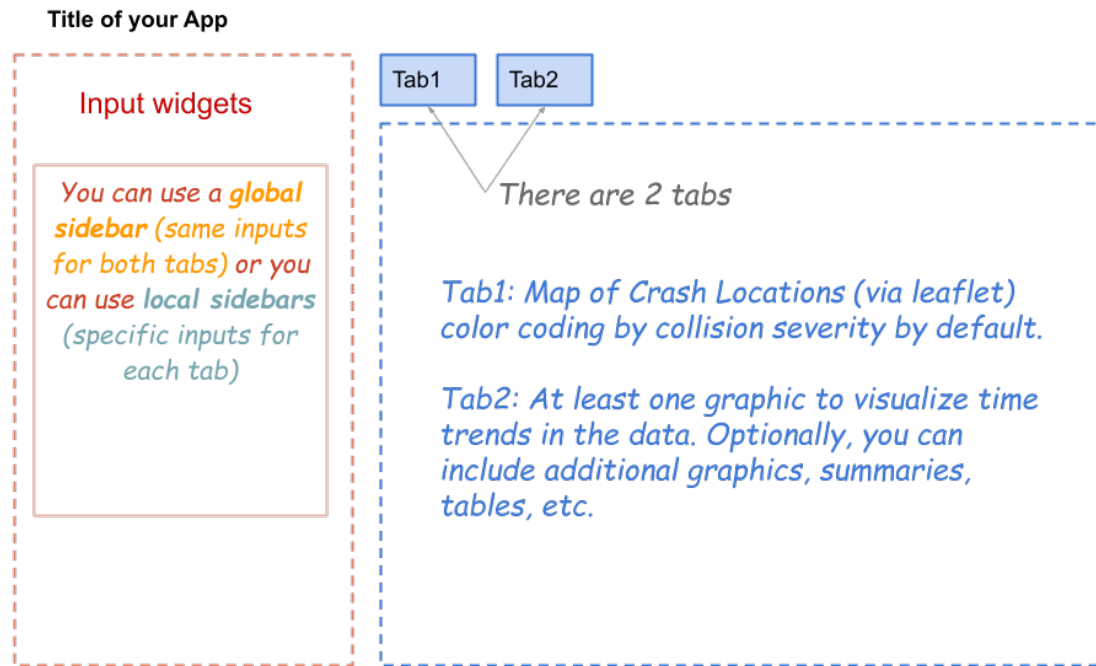


Figure 1: Layout of shiny dashboard provided in the template files.

2.4 Input widgets affecting map in Tab 1

Regardless of whether you decide to use a global sidebar or a local sidebar, the leaflet map should react to the inputs listed below. Optionally, you can include more inputs in addition to the required ones.

- 1) `selectInput()` or `selectizeInput()` to select a **county**. Default: Alameda county.
- 2) `selectInput()` or `selectizeInput()` to select a **violation category** (PCF_VIOL_CATEGORY). Default: "DUI" violation .
- 3) `checkboxGroupInput()` to choose one or more **types of collisions** (TYPE_OF_COLLISION). All types chosen by default.
- 4) `radioButtons()` to decide if locations should be color coded **by severity** (COLLISION_SEVERITY). Default: "yes". **BTW: If the user chooses to color code, then the map should include a legend for the meaning of the colors!**
- 5) `dateRangeInput()` to specify a certain **range of dates**. Default chosen values from "2019-01-01" to "2019-12-31".
- 6) `selectInput()` or `selectizeInput()` to change a **tile provider** for leaflet. You can find many examples here: <https://leaflet-extras.github.io/leaflet-providers/preview/>. See the storm tracks demo dashboard to get an idea of the various types of tile providers.

As we said, you are allowed to include more widgets that enhance the user experience, or that provide deeper insight into the exploratory analysis of crash locations.

2.5 Input widgets affecting output(s) in Tab 2

The content of Tab 2 is highly open-ended. The primary purpose of this tab is to display at least one graphic to visualize some sort of time trend in the data. For example: crashes by month, or crashes by day of the week, or crashes by hour of day, etc.

If you approached **Seasonality Trends** or **Weekend -vs- Weekday Collisions** or **The “Rush Hour” Effect** in your project 4, you can definitely take advantage of these analyses for the output(s) in tab 2.

Optionally, you can add more graphics, numeric summaries, value boxes, tables, or other output that provide deeper insight into the exploratory analysis of crashes across California or in a single county.